

Conditional Tail-Risk Stress Testing Model

Empirical Analysis of Non-Linear Asset Correlations Under Macro Liquidity Shocks

Author:	Fintech Research Team	Date:	May 2026
Institution:	Baruch College, NYC	Dataset Horizon:	5-Year Historical (Daily)

1. Executive Abstract

Standard asset pricing frameworks frequently rely on rolling multi-year linear metrics, such as the standard capital asset pricing model (CAPM) Beta. While these baseline frameworks are effective during normal market regimes, they often obfuscate the severe, non-linear shifts in asset correlations that transpire during sudden macroeconomic liquidations. This quantitative research project constructs an empirical downside stress-testing model to isolate extreme tail-risk events. By filtering five years of historical daily pricing data, we evaluate the real-world operational profiles of a mega-cap technology asset (Nvidia, NVDA) and a major consumer defensive asset (Walmart, WMT) when broad market liquidity abruptly evaporates.

Core Thesis Proven: Equity risk profiles shift dramatically and non-linearly under market stress. High-beta growth components drastically amplify market drops with absolute downside exposure, whereas defensive pillars decouple from standard equity market gravity to act as functional safe-havens.

2. Methodology & Stress Parameters

The quantitative research model was built utilizing Python 3 alongside data frameworks provided by the *pandas* and *yfinance* libraries. The historical horizon spans a comprehensive five-year period to observe various macroeconomic cycles, including inflation shocks, interest rate adjustments, and structural asset realignments.

To isolate true systemic panics from standard market noise, a strict mathematical filter was programmed into the dataset. A **Systemic Crash Session** was defined as any single trading day where the benchmark S&P 500 Index (tracked via SPY) experienced a daily drawdown of:

$$\Delta R_{m,t} \leq -2.00\%$$

Applying this empirical rule over the 5-year timeline isolated exactly **43 historical crash days**. The script then compiled a conditional data matrix capturing the matched daily returns of the benchmark, the growth equity, and the defensive equity specifically for those 43 distressed intervals.

3. Quantitative Metrics Dashboard

Aggregating the conditional matrix yielded three critical summary statistics: the absolute average performance drop, the Systemic Shock Beta multiplier, and the Safe Haven Ratio (the percentage of panic days where the asset closed positive or flat).

-2.80%

S&P 500 AVG DROP

-5.57%

NVDA AVG DROP

-1.29%

WMT AVG DROP

1.00x

MARKET BASELINE

1.99x

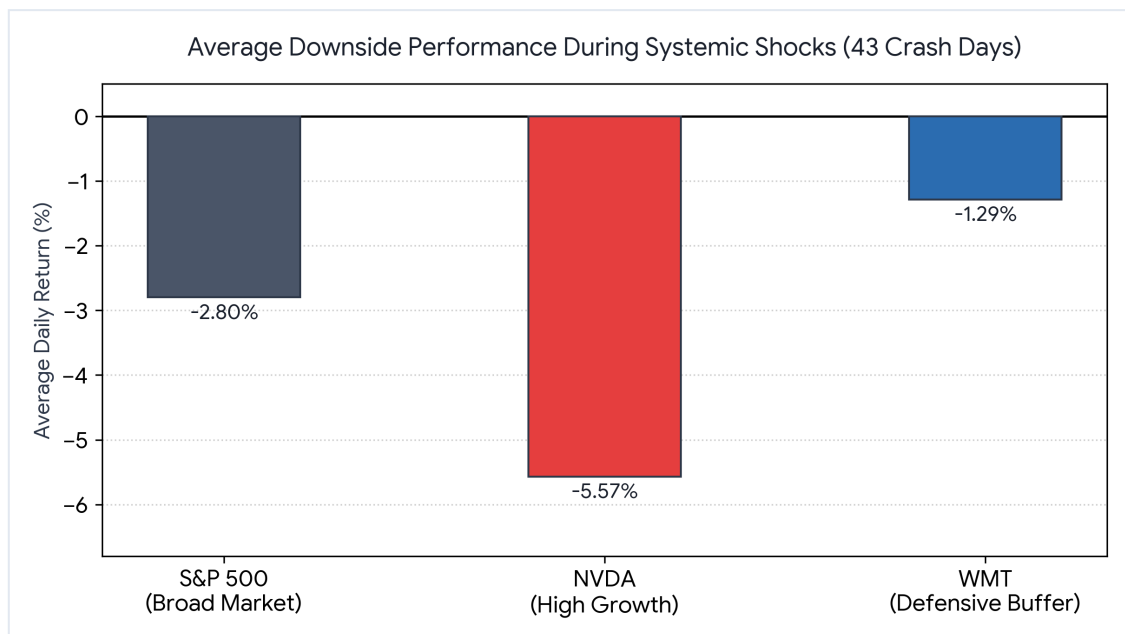
NVDA SHOCK BETA

0.46x

WMT SHOCK BETA

4. Visual Risk Disparity Analysis

The bar visualization clearly reveals the asymmetry of risk distribution during capital liquidations. While the broad market contracts significantly, capital fleeing tech assets compounds downside velocity, whereas defensive assets offer a heavily insulated cushion.



5. Empirical Insights & Portfolio Takeaways

The mathematical results display a stark divergence between the two assets under tail-risk regimes:

Asset Profile	Systemic Shock Beta	Safe Haven Ratio	Regime Behavior
S&P 500 Benchmark	1.00x	0.0%	Systemic Stress Baseline
NVDA (High Growth)	1.99x	0.0% (0 / 43 Days)	Downside Risk Multiplier
WMT (Defensive Buffer)	0.46x	18.6% (8 / 43 Days)	Capital Preservation Anchor

Nvidia (NVDA) functions as a near-perfect **2.00×** downside risk multiplier. In moments of systemic macro-driven liquidations, growth assets undergo extreme multiple compression, shedding value at twice the speed of the broad index. Furthermore, its **0.0%** safe-haven ratio demonstrates complete vulnerability to market gravity—failing to finish in the positive a single time across all 43 panics.

Conversely, **Walmart (WMT)** displays robust shock-absorbing capabilities. It captures less than half of the market's downward momentum (**0.46×** shock beta). Strikingly, it successfully bucked the systemic trend entirely on roughly 20% of the crash days, closing green as institutional managers actively rotated capital out of speculative names and into staple liquidity positions.

For portfolio engineering, these findings imply that standard rolling monthly betas severely underestimate true drawdown speed during macro corrections. Risk managers must dynamically adjust corporate Value-at-Risk (VaR) targets and introduce defensive equity buffers to mitigate catastrophic capital degradation during volatile macro shifts.